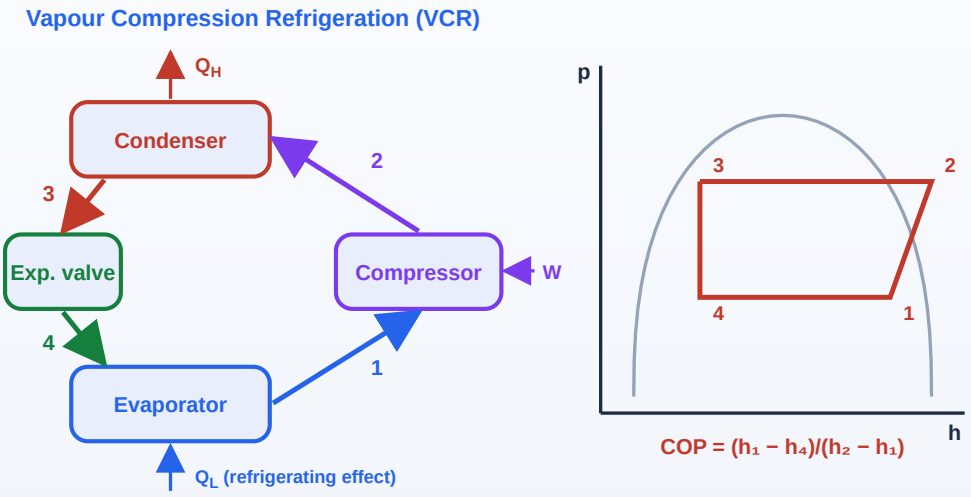
UPSSSC ALERT

UPSSSC JE

MECHANICAL REFRIGERATION & AIR COMPRESSION

Complete English Edition

Unit 7 · 5 Marks · 2026



UPSSSC JE (Mechanical) 2026: Part-1 has **5 questions (5 marks)** from **Refrigeration and Air Compression**. Syllabus: air compressors and their cycles; principles of refrigeration plants.

Each chapter contains key points, diagrams, formula table, detailed explanation, solved numericals, 30 one-liners, MCQs with explanations and SSC JE PYQs. Content follows standard BTEUP diploma textbooks (C. P. Arora, R. S. Khurmi, R. K. Rajput).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Air Compressors & their Cycles

Air Compressors & their Cycles – Basic Concept

An **air compressor** takes in atmospheric air and raises its pressure. Compressed air is used for pneumatic tools (drills, hammers), spray painting, tyre inflation, brakes (trains/buses), gas turbines, supercharging and refrigeration. Work must be supplied to drive a compressor.

Understand in Detail — Air Compressors & their Cycles

1.1 What is an Air Compressor and why do we need it?

When you fill air in a bicycle tyre with a hand pump, you are actually using a small air compressor. Pushing the handle down reduces the volume of air inside the cylinder, its pressure rises, and when this pressure exceeds the pressure inside the tyre the valve opens and air enters the tyre. An industrial compressor does exactly the same job — only on a larger scale and driven by a motor or an engine.

Definition: An air compressor is a machine that takes in air from the atmosphere, does external work on it and delivers it at a higher pressure. That is, a compressor is a **work-absorbing machine** — it does not produce energy; it takes energy from a motor and increases the pressure energy of air. That is why it is sometimes called an 'engine running in reverse'.

Uses of compressed air: pneumatic tools (drills, riveting hammers), spray painting, supplying air to blast furnaces, gas turbines, starting diesel engines, air brakes of trains and trucks, ventilation of mines, and in refrigeration and liquefaction of gases. In mines an electric spark can cause a fire, so tools driven by compressed air are considered the safest there.

Classification: (1) **By working principle** — reciprocating (piston-cylinder) and rotary (vane, Roots blower, screw, centrifugal, axial). (2) **By action** — single-acting (compression on one side of the piston only) and double-acting (compression on both sides, so delivery twice per revolution). (3) **By stages** — single-stage and multi-stage. (4) **By pressure** — low (up to 10 bar), medium (10–80 bar) and high (above 80 bar).

Positive displacement vs dynamic: reciprocating, vane, Roots and screw compressors are **positive displacement** — a fixed quantity of air is trapped in a closed space and its volume is reduced. Centrifugal and axial compressors are **dynamic** — air is first given a high velocity and then the velocity is converted into pressure in a diffuser. General rule: **small quantity + high pressure → reciprocating; large quantity + low pressure → rotary/centrifugal/axial.**

> **Note:** In a Roots blower there is no internal compression — the pressure rises only when the air meets the high-pressure air on the delivery side (by back-flow). Hence its efficiency is low. A vane compressor has partial internal compression.

1.2 Construction and working of a reciprocating compressor

A reciprocating compressor looks almost like an engine — it has a cylinder, piston, connecting rod, crankshaft and two valves in the cylinder head: the **suction (inlet) valve** and the **delivery valve**. The difference is that in an engine the valves are opened by a cam, whereas compressor valves are **automatic plate valves** that open and close only due to pressure difference.

Suction stroke: as the piston moves outward, the cylinder volume increases and the pressure inside falls slightly below atmospheric. This pressure difference opens the suction valve and air enters. The delivery valve stays closed because the high receiver pressure acts on it.

Compression: as soon as the piston returns, the pressure inside rises and the suction valve closes. Now both valves are closed and the volume of air keeps decreasing — the pressure keeps rising. This process follows $p\,V^n$ = constant.

Delivery: as soon as the cylinder pressure becomes slightly more than the receiver pressure, the delivery valve opens and during the rest of the stroke the air is pushed at constant pressure into the receiver tank. The receiver acts as a store and smooths out flow pulsations.

Clearance volume (V_c): so that the piston never hits the cylinder head and there is room for the valves, a small volume remains even at the end of the stroke. This is the clearance volume. The high-pressure air left in it cannot be delivered — we will see its effect on volumetric efficiency later.

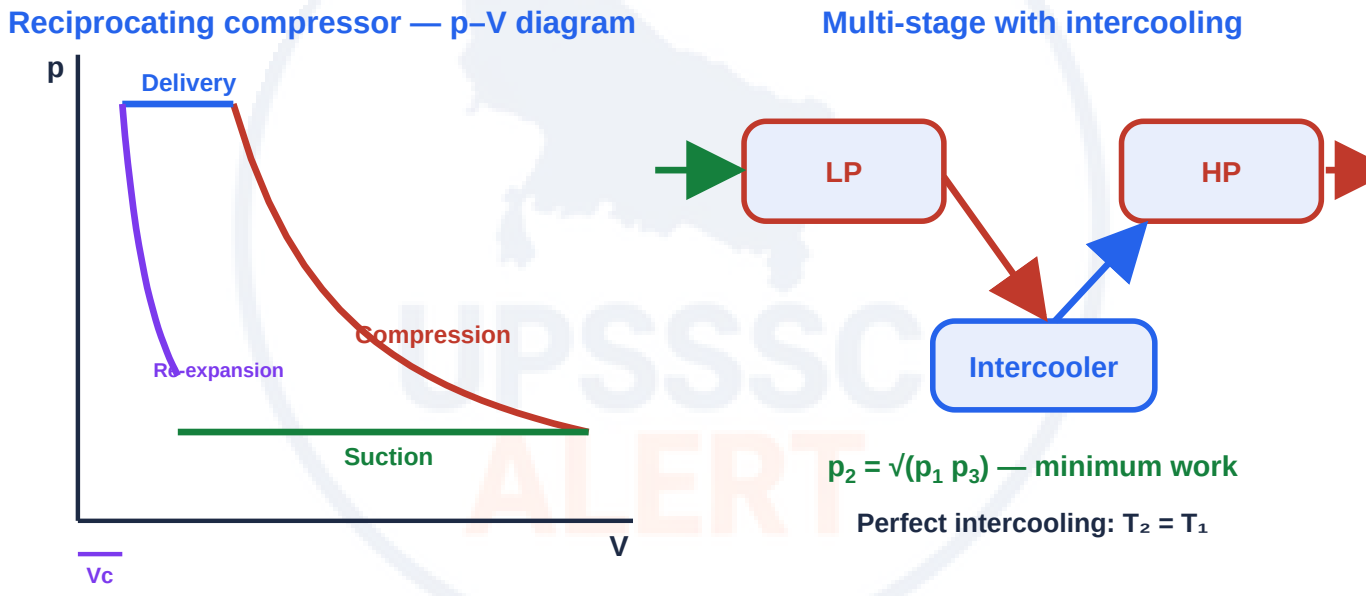


Fig 7.1: Reciprocating air compressor and two-stage compression

1.3 Learn to read the p–V diagram — the area is the work

Look carefully at the left diagram in the figure above. The horizontal axis is volume (V) and the vertical axis is pressure (p). Ignoring clearance for now, the cycle has four lines: (1) **suction** — a horizontal line at pressure p_1, volume 0 to V_1; (2) **compression** — a curve from p_1V_1 to p_2V_2; (3) **delivery** — a horizontal line at pressure p_2, volume V_2 to 0; (4) the drop from p_2 to p_1 when the valve opens.

Why is the area = work? The unit of pressure × volume ($N/m^2 \times m^3 = N \cdot m = J$) is the unit of energy. The area enclosed by the cycle is the net work the motor must do on the air per cycle. It is the area seen from the **pressure axis**, not the volume axis, so the work of a compressor (an open, flow system) is **$W = \int V\,dp$**, not $\int p\,dV$.

Which compression needs the least work? From the same p_1 to p_2 compression can happen in three ways — isothermal ($pV = C$, $n = 1$), polytropic ($pV^n = C$, $1 < n < \gamma$) and adiabatic ($pV^\gamma = C$). Air heats up during compression. If the heat is removed (isothermal), the air stays cool and occupies less volume, so the curve stays to the left and the enclosed area is the least. In adiabatic compression no heat is removed, the air is hottest and most expanded, so the area is the largest. Hence **$W_{iso} < W_{poly} < W_{adia}$**.

That is why **fins** or a **water jacket** are provided on the compressor cylinder — so that compression comes as close to isothermal as possible, n reduces and work is saved. In practice $n \approx 1.2 – 1.35$.

1.4 Deriving the work formula (without clearance)

Now we find the area mathematically. Area of cycle = area under delivery line − area under suction line + area under compression curve (measured from the V-axis).

Derivation: Polytropic compression $pV^n = C$

1. Work done during delivery = p_2V_2 ; work done by air on piston during suction = p_1V_1
2. Work done during compression = $(p_2V_2 - p_1V_1)/(n - 1)$
3. Total $W = p_2V_2 - p_1V_1 + (p_2V_2 - p_1V_1)/(n - 1) =$ **$[n/(n - 1)]\,(p_2V_2 - p_1V_1)$**
4. Take p_1V_1 out: $W = [n/(n - 1)]\,p_1V_1\,[(p_2V_2)/(p_1V_1) - 1]$
5. From $pV^n = C$, $V_2/V_1 = (p_1/p_2)^{1/n}$, so $(p_2V_2)/(p_1V_1) = (p_2/p_1)^{(n-1)/n}$
6. **$W = [n/(n - 1)]\,p_1V_1\,[(p_2/p_1)^{(n-1)/n} - 1]$** ; putting $p_1V_1 = mRT_1$ gives $W = [n/(n - 1)]\,mR(T_2 - T_1)$

Derivation: Isothermal compression $pV = C$

1. Here $p_2V_2 = p_1V_1$, so the delivery and suction works cancel out
2. Only the compression work remains = $p_1V_1 \ln(V_1/V_2)$
3. **$W_{iso} = p_1V_1 \ln(p_2/p_1) = mRT_1 \ln(p_2/p_1)$**

Delivery temperature: $T_2/T_1 = (p_2/p_1)^{(n-1)/n}$. The higher the pressure ratio, the higher the delivery temperature. At very high temperature the lubricating oil may burn (carbonisation) and there is even a risk of explosion — this is one reason the pressure ratio of a single-stage compressor is kept limited (usually up to 5–7).

Power: if V_1 is the volume sucked per minute, W comes in kJ/min; divide by 60 for kW. A single-acting compressor delivers once per revolution (N rev/min → N deliveries), a double-acting one 2N.

Example: A single-stage compressor takes 1 m³/min of air at 1 bar, 27°C and delivers it at 7 bar. Taking n = 1.3 find (i) power, (ii) isothermal power, (iii) isothermal efficiency, (iv) delivery temperature.

Solution:

1. Pressure ratio r = 7/1 = 7 ; (n−1)/n = 0.3/1.3 = 0.2308 ; n/(n−1) = 4.333
2. $7^{0.2308}$ = 1.567 ; p_1V_1 = 100 kPa × 1 m³/min = 100 kJ/min
3. W = 4.333 × 100 × (1.567 − 1) = 245.7 kJ/min → P = 245.7/60 = **4.09 kW**
4. W_{iso} = 100 × ln 7 = 100 × 1.946 = 194.6 kJ/min → **3.24 kW**
5. η_{iso} = 194.6/245.7 = **0.792 ≈ 79.2%**
6. T_2 = 300 × 1.567 = 470 K = **197°C** — see, at just 7 bar the air got heated to nearly 200°C!

1.5 Clearance volume and volumetric efficiency — full understanding

At the end of the delivery stroke air at pressure p_2 remains in the clearance volume V_c. When the piston returns, the suction valve does not open immediately — first this trapped air **re-expands** and its pressure falls from p_2 to p_1. While this expansion is going on, that part of the piston travel is 'wasted'; fresh air enters only during the remaining stroke.

Volumetric efficiency (η_v) = actual volume of air sucked (at suction conditions) ÷ swept volume (V_s). The larger the clearance and the higher the pressure ratio, the longer the re-expansion and the lower η_v.

Derivation: Formula for η_v

1. Clearance ratio C = V_c/V_s ; total volume at start of stroke = $V_c + V_s$
2. After re-expansion the volume of clearance air $V_4 = V_c(p_2/p_1)^{1/n}$
3. Volume sucked = $(V_c + V_s) - V_4 = V_s + V_c - V_c(p_2/p_1)^{1/n}$
4. Dividing by V_s: $\boldsymbol{\eta_v = 1 + C - C\,(p_2/p_1)^{1/n}}$
5. $\eta_v = 0$ when $(p_2/p_1)^{1/n} = (1 + C)/C$ — this is the **maximum possible pressure ratio**; beyond it the compressor delivers no air at all

Interesting fact: even with clearance the **work per kg of air does not change** — because the clearance air takes work in compression and returns nearly the same work in re-expansion (like a spring). The real loss due to clearance is that **a cylinder of the same size delivers less air**, so the machine has to be made bigger.

Free Air Delivery (FAD): the volume of air delivered by the compressor measured at atmospheric (free) pressure and temperature. A compressor's capacity is always stated as FAD (m³/min). **Isothermal efficiency** = isothermal work ÷ actual indicated work — it tells how close our compression is to the ideal (isothermal).

Example: With clearance ratio 5%, pressure ratio 7 and n = 1.3, find the volumetric efficiency and the maximum possible pressure ratio.

Solution:

1. $(p_2/p_1)^{1/n} = 7^{1/1.3} = 7^{0.769} = 4.468$
2. $\eta_v = 1 + 0.05 - 0.05 \times 4.468 = 1.05 - 0.223 =$ **$0.827 \approx 82.7\%$**
3. Maximum ratio: $(p_2/p_1)^{1/1.3} = 1.05/0.05 = 21 \rightarrow p_2/p_1 = 21^{1.3} =$ **≈ 52.4**

1.6 Multi-stage compression and intercooling — why and how?

If we compress from 1 bar to 50 bar in a single cylinder there are three problems: (1) the delivery temperature becomes very high, (2) the volumetric efficiency falls badly (close to zero), (3) much more work is needed. The solution is **to divide the compression into several stages** and cool the air between stages.

Arrangement: air is first compressed in the **low-pressure (LP) cylinder** from p_1 to an intermediate pressure p_2. Then it passes through an **intercooler** — a water-cooled tube heat exchanger — where its temperature is reduced. The cooled air is then compressed in the **high-pressure (HP) cylinder** to p_3. The HP cylinder is smaller because cool, compressed air has a smaller volume.

Perfect intercooling: when the intercooler cools the air back exactly to the initial temperature T_1. Then the starting point of the second stage lies on the isothermal curve itself. The 'cut-off' portion between the two stages on the p–V diagram is the **work saved** — that is why the more the stages, the closer the compression comes to isothermal.

Derivation: Intermediate pressure for minimum work

1. With perfect intercooling both stages have the same $p_1V_1 = mRT_1$, so $W = [n/(n-1)]\ mRT_1\ [(p_2/p_1)^k + (p_3/p_2)^k - 2]$, where $k = (n-1)/n$
2. p_1 and p_3 are fixed; to minimise W put $dW/dp_2 = 0$
3. $k\ p_2^{k-1}/p_1^k - k\ p_3^k\ p_2^{-k-1} = 0 \rightarrow p_2^{2k} = (p_1p_3)^k$
4. **$p_2 = \sqrt{(p_1\ p_3)}$**, i.e. $p_2/p_1 = p_3/p_2$ — **both stages have equal pressure ratio**
5. Result: both stages do **equal work**, have equal delivery temperatures and nearly equal heat rejected
6. For N stages: ratio per stage $= (p_{final}/p_1)^{1/N}$; total $W = N\ [n/(n-1)]\ p_1V_1\ [(p_{final}/p_1)^{(n-1)/(Nn)} - 1]$

Advantages of multi-staging at a glance: less work (saving electricity), lower delivery temperature (lubrication safe), higher volumetric efficiency (each stage has a small pressure ratio), better mechanical balance, and the HP cylinder can be small and strong. Disadvantage — the machine is complex and costly.

Example: 1 m^3 of air is to be compressed from 1 bar to 16 bar (n = 1.3). Find the work for (i) single-stage and (ii) two-stage compression with perfect intercooling, and the saving.

Solution:

1. Single stage: $16^{0.2308} = 1.896 \rightarrow W = 4.333 \times 100 \times 0.896 =$ **388.3 kJ**
2. Two stage: $p_2 = \sqrt{(1 \times 16)} = 4$ bar, ratio of each stage 4 ; $4^{0.2308} = 1.377$
3. $W = 2 \times 4.333 \times 100 \times 0.377 =$ **326.7 kJ**
4. Saving = 388.3 − 326.7 = 61.6 kJ, i.e. **about 15.9%** less work
5. Delivery temperature: single stage $300 \times 1.896 = 569$ K (296°C), two stage $300 \times 1.377 = 413$ K (140°C) — what a big difference!

Note: Exams often ask — 'intermediate pressure for minimum work with perfect intercooling?' Answer: **geometric mean $\sqrt{(p_1 p_3)}$**, not arithmetic mean $(p_1+p_3)/2$. Example: 1 to 9 bar → 3 bar, not 5 bar.

1.7 Rotary compressors — vane, Roots, screw, centrifugal and axial

Vane compressor: a rotor is mounted **eccentrically** in a cylindrical casing. Sliding vanes fitted in rotor slots move outward by centrifugal force and stay pressed against the casing wall. Because of the eccentricity the space between two vanes first increases (suction) and then decreases (compression) as it rotates. Pressure: up to about 8 bar.

Roots blower: two- or three-lobe rotors rotate in opposite directions and carry air along the casing wall. No internal compression; pressure ratio up to about 2. Uses — engine superchargers, air supply.

Screw compressor: two helical rotors, male and female, mesh with each other. Air trapped between them moves axially as the rotors turn and its volume continuously decreases. Low vibration, continuous flow — the most popular in factories today.

Centrifugal compressor: air enters the eye of the impeller axially, the fast-rotating impeller throws it out radially and gives it velocity; then in the **diffuser** the velocity falls and the pressure rises. Roughly half the pressure rise occurs in the impeller and half in the diffuser. **Axial compressor:** air flows parallel to the axis; there are many rows of rotating blades (rotor) and fixed blades (stator) — each row raises the pressure a little. It is used in gas turbines and jet engines because it handles a very large flow in a small frontal area.

Surging and choking: in centrifugal/axial compressors, if the flow becomes too small, air starts flowing backward and the flow oscillates violently — this is surging (noise, vibration, damage). If the flow becomes too large and somewhere the velocity reaches the speed of sound, no further increase in flow is possible — this is choking.

1. Key Points

- **Classification:** reciprocating (piston–cylinder; low flow, high pressure) and rotary (Roots blower, vane, screw, centrifugal, axial; high flow, lower pressure ratio).

- **Types of compression:** isothermal ($pV = C$) — minimum work; polytropic ($pV^n = C$, $1 < n < \gamma$) — practical; adiabatic ($pV^\gamma = C$) — maximum work.
- **Isothermal work:** $W = p_1V_1 \ln(p_2/p_1)$. **Polytropic work:** $W = [n/(n-1)]\, p_1V_1\, [(p_2/p_1)^{(n-1)/n} - 1]$.
- **Isothermal efficiency:** η_{iso} = isothermal work ÷ actual work. Cooling (water jacket, fins) is needed to reduce work.
- **Clearance volume:** compressed air left at TDC re-expands before fresh air enters — this lowers volumetric efficiency but does not change work per kg.
- **Volumetric efficiency:** $\eta_v = 1 + C - C(p_2/p_1)^{1/n}$, where $C = V_c/V_s$. η_v falls as pressure ratio rises.
- **Multi-stage compression:** two or more stages with an intercooler in between for high pressure; reduces work, improves volumetric efficiency and approaches isothermal compression.
- **Perfect intercooling and minimum work:** for two stages the intermediate pressure $p_2 = \sqrt{(p_1p_3)}$; then work in both stages is equal. For N stages the pressure ratio per stage = $(p_{N+1}/p_1)^{1/N}$.
- **Free air delivered (FAD):** volume flow of compressed air reduced to atmospheric conditions (usually 1.013 bar, 15°C).
- **Centrifugal compressor:** the impeller accelerates air radially and the diffuser converts velocity into pressure; used in gas turbines, jet engines and superchargers.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Isothermal work	$W = p_1V_1 \ln(p_2/p_1)$
Polytropic work	$W = [n/(n-1)]\, p_1V_1\, [(p_2/p_1)^{(n-1)/n} - 1]$
Adiabatic work	$W = [\gamma/(\gamma-1)]\, p_1V_1\, [(p_2/p_1)^{(\gamma-1)/\gamma} - 1]$
Isothermal efficiency	$\eta_{iso} = W_{iso}/W_{actual}$
Volumetric efficiency	$\eta_v = 1 + C - C(p_2/p_1)^{1/n}$
Two-stage intermediate pressure	$p_2 = \sqrt{(p_1\, p_3)}$
N stages — ratio per stage	$(p_{final}/p_1)^{1/N}$
N-stage minimum work	$W = N[n/(n-1)]p_1V_1[(p_f/p_1)^{(n-1)/(Nn)} - 1]$
Delivery temperature	$T_2 = T_1\, (p_2/p_1)^{(n-1)/n}$
Indicated power	$P = W \times N/60$ (single acting)

3. Detailed Explanation

- **Why isothermal is ideal:** on the p–V diagram the isothermal curve is the least steep, so the area (work) is minimum; in adiabatic compression the rise in temperature keeps volume larger, needing more work.
- **Benefit of intercooler:** cooling air to the initial temperature after the first stage reduces volume entering the second stage, saving work; it also prevents lubricating oil from burning at high

temperature.

- **Single and double acting:** single acting compresses on one side of the piston; double acting on both sides — two deliveries per revolution.
- **Rotary compressors:** the Roots blower has no internal compression (back-flow); the vane compressor has internal compression; axial compressors handle the largest flows.
- **Effect of clearance on indicator diagram:** larger clearance reduces the volume drawn in, but work per kg is unchanged because the re-expanding clearance air returns its work.

4. Comparison Table — Reciprocating vs Rotary Compressor

Basis	Reciprocating	Rotary
Flow	Low	High
Pressure ratio	High (multi-stage)	Low per stage
Speed	Low	High
Delivery	Pulsating	Continuous, smooth
Lubrication & air purity	Oil may mix	Clean air

4. Comparison Table — Isothermal, Polytropic and Adiabatic Compression

Basis	Isothermal	Polytropic	Adiabatic
Law	$pV = C$	$pV^n = C$	$pV^\gamma = C$
Work	Minimum	Intermediate	Maximum
Delivery temperature	Equal to T_1	Moderate	Highest

5. Solved Numerical Problems

Example 1. 1 m³ of air is compressed isothermally from 1 bar to 7 bar. Find the work.

Solution:

1. $W = 100 \times 1 \times \ln 7 = 100 \times 1.946 =$ **194.6 kJ**

Example 2. Solve the above for polytropic compression with n = 1.3 and find the isothermal efficiency.

Solution:

1. $(7)^{0.3/1.3} = 7^{0.2308} = 1.567$
2. $W = (1.3/0.3) \times 100 \times (1.567 - 1) = 4.333 \times 56.7 =$ **245.7 kJ**
3. $\eta_{iso} = 194.6/245.7 =$ **79.2%**

Example 3. A two-stage compressor compresses from 1 bar to 16 bar. Find the intermediate pressure for minimum work.

Solution:

1. $p_2 = \sqrt{(1 \times 16)} =$ **4 bar**

Example 4. Clearance ratio C = 0.05, pressure ratio 8, n = 1.3. Find the volumetric efficiency.

Solution:

1. $8^{1/1.3} = 8^{0.769} = 4.95$
2. $\eta_V = 1 + 0.05 - 0.05 \times 4.95 =$ **80.2%**

Example 5. Air at 27°C is compressed from 1 to 8 bar with n = 1.3. Find the delivery temperature.

Solution:

1. $T_2 = 300 \times 8^{0.3/1.3} = 300 \times 1.616 =$ **484.8 K (211.8°C)**

6. Frequently Asked Facts

- Isothermal compression needs minimum work, adiabatic the maximum.
- Two-stage minimum work: $p_2 = \sqrt{(p_1 p_3)}$, equal work in both stages.
- Clearance lowers volumetric efficiency, not work per kg.
- Isothermal efficiency = isothermal work ÷ actual work.

7. Note (Confusing Facts)

- Volumetric efficiency decreases (not increases) as pressure ratio rises.
- There is no internal compression in a Roots blower.
- Perfect intercooling means the second-stage inlet temperature equals the first-stage inlet temperature.

Remember: One line: the lower n, the lower the work (n = 1 isothermal). Assertion–Reason: (A) Multi-stage compression is used for high pressure. (R) Intercooling reduces work and improves volumetric efficiency. — Both true; R explains A.

30 Most Important One-Liner Facts — Air Compressors & their Cycles

1. Compression needing least work?

— Isothermal.

2. Compression needing most work?

— Adiabatic.

3. Practical compression?

— Polytropic ($pV^n = C$).

4. Isothermal work?

— $p_1V_1 \ln(p_2/p_1)$.

5. Isothermal efficiency?

— Isothermal ÷ actual work.

6. Volumetric efficiency formula?

— $1 + C - C(p_2/p_1)^{1/n}$.

7. Clearance ratio?

— V_c/V_s.

8. η_V with more clearance?

— Decreases.

9. η_v with higher pressure ratio?

— Decreases.

10. Clearance effect on work per kg?

— None.

11. Device between stages?

— Intercooler.

12. Two-stage intermediate pressure?

— $\sqrt{(p_1 p_3)}$.

13. Stage works at minimum?

— Equal.

14. Ratio per stage for N stages?

— $(p_f/p_1)^{1/N}$.

15. Perfect intercooling?

— $T_2 = T_1$.

16. Benefit of multi-staging?

— Less work, higher η_v.

17. FAD stands for?

— Free Air Delivered.

18. No internal compression?

— Roots blower.

19. Compressor in jet engine?

— Axial / centrifugal.

20. Pressure-raising part of centrifugal?

— Diffuser.

21. Low flow, high pressure?

— Reciprocating.

22. High flow, low pressure?

— Rotary.

23. Means of compressor cooling?

— Water jacket, fins.

24. Compression on both sides?

— Double acting.

25. Delivery temperature?

— $T_1(p_2/p_1)^{(n-1)/n}$.

26. Extra benefit of intercooler?

— Lube oil does not burn.

27. Clearance air process on p–V?

— Re-expansion.

28. Use of compressed air?

— Pneumatic tools, brakes.

29. 1 to 16 bar two-stage — p_2?

— 4 bar.

30. γ of air?

— 1.4.

Practice MCQs — Air Compressors & their Cycles

Q1. Minimum work in air compression is for:

(a) Adiabatic
(b) Isothermal
(c) Polytropic
(d) Isentropic

✔ **Answer: (b) Isothermal**

Explanation: pV = C.

Q2. With increased clearance volume, volumetric efficiency:

(a) Increases
(b) Decreases
(c) Constant
(d) Zero

✔ **Answer: (b) Decreases**

Explanation: Re-expansion reduces intake.

Q3. Intermediate pressure for minimum work in a two-stage compressor:

(a) $(p_1 + p_3)/2$ (b) $\sqrt{(p_1p_3)}$

(c) p_1p_3 (d) $p_3 - p_1$

✔ **Answer: (b) $\sqrt{(p_1p_3)}$**

Explanation: Geometric mean.

Q4. Isothermal efficiency =

(a) Actual ÷ isothermal work (b) Isothermal ÷ actual work

(c) Adiabatic ÷ isothermal (d) FAD ÷ swept

✔ **Answer: (b) Isothermal ÷ actual work**

Explanation: Ideal ÷ actual.

Q5. Two-stage from 1 to 9 bar — intermediate pressure:

(a) 3 bar (b) 4.5 bar

(c) 5 bar (d) 9 bar

✔ **Answer: (a) 3 bar**

Explanation: √9.

Q6. Purpose of intercooling:

(a) Increase work (b) Reduce work

(c) Reduce pressure (d) Reduce flow

✔ **Answer: (b) Reduce work**

Explanation: Approaches isothermal.

Q7. In a Roots blower:

(a) Internal compression occurs (b) No internal compression

(c) There is a piston (d) There are valves

✔ **Answer: (b) No internal compression**

Explanation: Pressure rises by back-flow.

Q8. Volumetric efficiency as pressure ratio increases:

(a) Increases (b) Decreases

(c) Constant (d) First rises then falls

✔ **Answer: (b) Decreases**

Explanation: $(p_2/p_1)^{1/n}$ increases.

Q9. Effect of clearance on work per kg:

(a) Increases (b) Decreases

(c) No effect (d) Doubles

✔ **Answer: (c) No effect**

Explanation: Re-expansion returns work.

Q10. 1 m^3 air from 1 to e (2.718) bar isothermally — work:

(a) 100 kJ
(b) 271.8 kJ
(c) 50 kJ
(d) 1 kJ

✔ **Answer: (a) 100 kJ**

Explanation: 100 × ln e = 100.

Q11. Perfect intercooling means:

(a) Second-stage inlet temp = initial temp
(b) Equal pressure
(c) No cooling
(d) Adiabatic compression

✔ **Answer: (a) Second-stage inlet temp = initial temp**

Explanation: $T_2 = T_1$.

Q12. Compressor suited to very high flow:

(a) Reciprocating
(b) Axial
(c) Vane
(d) Roots

✔ **Answer: (b) Axial**

Explanation: Axial.

Q13. Pressure ratio per stage in an N-stage compressor:

(a) $p_f/(Np_1)$
(b) $(p_f/p_1)^{1/N}$
(c) $N(p_f/p_1)$
(d) $(p_f/p_1)^N$

✔ **Answer: (b) $(p_f/p_1)^{1/N}$**

Explanation: Equal ratios.

Q14. In a centrifugal compressor velocity is converted to pressure in the:

(a) Impeller
(b) Diffuser
(c) Shaft
(d) Inlet

✔ **Answer: (b) Diffuser**

Explanation: Diffuser.

Q15. C = 0.05, $p_2/p_1 = 8$, n = 1.3 — η_v about:

(a) 70%
(b) 80%
(c) 90%
(d) 95%

✔ **Answer: (b) 80%**

Explanation: 1.05 − 0.05 × 4.95.

SSC JE / UPSSSC JE Previous Year Questions — Air Compressors & their Cycles

Q1. Isothermal efficiency of an air compressor is highest when compression is: SSC JE (Previous Year)

(a) Adiabatic
(b) Close to isothermal
(c) Isentropic
(d) Constant volume

✔ **Answer: (b) Close to isothermal**

Explanation: Good cooling.

Q2. Advantage of multi-stage compression: SSC JE (Previous Year)

(a) Reduction in work
(b) Reduction in volumetric efficiency
(c) Higher temperature
(d) More work

✔ **Answer: (a) Reduction in work**

Explanation: Intercooling.

Q3. Clearance volume in a reciprocating compressor reduces its: SSC JE (Previous Year)

(a) Volumetric efficiency
(b) Isothermal efficiency
(c) Mechanical efficiency
(d) Pressure

✔ **Answer: (a) Volumetric efficiency**

Explanation: Less intake volume.

Q4. For minimum work in a two-stage compressor: SSC JE (Previous Year)

(a) Equal pressure ratio in both stages
(b) Higher in first stage
(c) Higher in second stage
(d) No intercooler

✔ **Answer: (a) Equal pressure ratio in both stages**

Explanation: $p_2/p_1 = p_3/p_2$.

Q5. Purpose of cooling in a compressor: SSC JE (Previous Year)

(a) Reduce work
(b) Increase work
(c) Reduce pressure
(d) Increase flow

✔ **Answer: (a) Reduce work**

Explanation: Towards isothermal.

Chapter 2: Principles of Refrigeration Plants

Principles of Refrigeration Plants – Basic Concept

Refrigeration is the process of removing heat from a space or substance to bring and keep its temperature below the surroundings. By the second law (Clausius) this is impossible without external work. Domestic refrigerators, air conditioners, cold storages, ice plants and water coolers are examples.

Understand in Detail — Principles of Refrigeration Plants

2.1 What is refrigeration — pumping heat 'uphill'

Heat naturally always flows from a hot body to a cold body — just as water flows downhill. The inside of a fridge is about 4°C and the kitchen 30°C, yet the fridge keeps removing heat from inside and throwing it into the warmer room air. This is like pumping heat 'uphill', and just as pumping water uphill needs a pump and electricity, moving heat from cold to hot needs **external work** — this is the **second law of thermodynamics (Clausius statement)**.

Definition: the process of bringing the temperature of a space or substance below that of its surroundings and maintaining it there is called **refrigeration**. The substance that carries this heat is called the **refrigerant**.

Unit of refrigeration — Ton of Refrigeration (TR): the rate of heat removal needed to convert 1 ton (2000 lb ≈ 907 kg) of water at 0°C into ice at 0°C in 24 hours is 1 TR. **1 TR = 210 kJ/min = 3.5 kW** (approx.). A '1.5 ton AC' at home means a capacity to remove about 5.25 kW of heat from the room.

Coefficient of Performance (COP): the 'mileage' of a refrigerator — **COP = refrigerating effect obtained ÷ work supplied = Q_L/W**. Unlike efficiency, COP can be more than 1 and is usually 2.5 – 5, because we don't 'create' heat, we only 'carry' it from one place to another.

Derivation: COP of a Carnot refrigerator — what is the limit?

1. Energy balance: $Q_H = Q_L + W$ (heat rejected = heat taken in + work)
2. $COP_R = Q_L/(Q_H - Q_L)$; for a reversible cycle $Q_H/Q_L = T_H/T_L$
3. $\mathbf{COP_{R,\ Carnot} = T_L/(T_H - T_L)}$; for a heat pump $\mathbf{COP_{HP} = T_H/(T_H - T_L) = COP_R + 1}$
4. Conclusion: the smaller the difference between the two temperatures, the higher the COP. That is why an AC consumes more electricity in peak summer (45°C outside)

Example: A Carnot refrigerator works between −13°C and 27°C. Find the COP and the power needed for 1 TR.

Solution:

1. T_L = 260 K, T_H = 300 K (always convert to kelvin!)
2. COP = 260/(300 − 260) = 260/40 = **6.5**
3. W = Q_L/COP = 3.5/6.5 = **0.54 kW** ; as a heat pump COP = 7.5

2.2 Vapour Compression Refrigeration (VCR) — the heart of a household fridge

You may have noticed that spirit or petrol on your hand feels cold — because while evaporating the liquid takes **latent heat** from your skin. The VCR system works on this very principle: a liquid (refrigerant) that boils at a very low temperature is boiled at low pressure, and while boiling it draws heat from inside the fridge.

Four main components and the cycle (see figure):

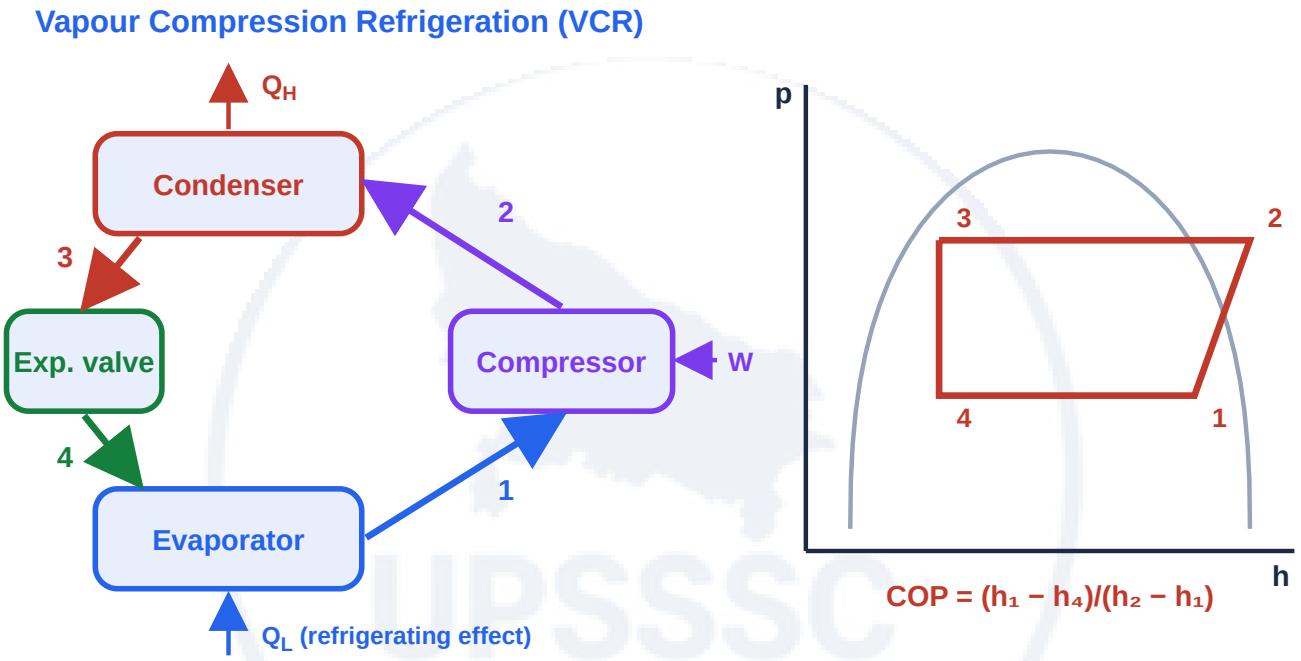


Fig 7.2: VCR cycle — components and p–h diagram

Derivation: COP from the p–h diagram

1. Refrigerating effect (per kg) = $h_1 - h_4$; compressor work = $h_2 - h_1$; heat in condenser = $h_2 - h_3$
2. Throttling gives $h_4 = h_3$
3. **COP = $(h_1 - h_4)/(h_2 - h_1)$**
4. Refrigerant mass flow: $\dot{m}$ = (capacity kW)/$(h_1 - h_4)$; compressor power = $\dot{m}$ $(h_2 - h_1)$

1 → 2 Compressor: the compressor compresses the low-pressure, low-temperature **dry saturated vapour** from the evaporator to high pressure. After compression (ideally isentropic) the vapour becomes

superheated and hot — its temperature becomes higher than the outside air. **This is the only place where work (electricity) is supplied.**

2 → 3 Condenser: the black wire-mesh tubes at the back of a fridge. The hot vapour gives heat (Q_H) to the outside air; first the superheat is removed, then it condenses at constant pressure into **high-pressure liquid**. That is why the back of a fridge feels warm.

3 → 4 Expansion valve / capillary tube: the high-pressure liquid passes through a very thin tube or valve where its pressure suddenly drops. This is **throttling — enthalpy remains constant ($h_3 = h_4$)**, there is no work or heat transfer. As the pressure drops, some liquid instantly flashes into vapour and the temperature of the mixture falls very low (e.g. −20°C).

4 → 1 Evaporator: the cold freezer plates. The cold, low-pressure wet mixture takes heat (Q_L) from the food inside the fridge, boils and becomes **dry saturated vapour** again, which goes to the compressor — and the cycle repeats. **$Q_L = h_1 - h_4$ is the refrigerating effect.**

Example: In a VCR system h_1 = 1450, h_2 = 1650, $h_3 = h_4$ = 350 kJ/kg (ammonia). For 10 TR capacity find COP, refrigerant flow and compressor power.

Solution:

1. Refrigerating effect = 1450 − 350 = 1100 kJ/kg ; work = 1650 − 1450 = 200 kJ/kg
2. COP = 1100/200 = **5.5**
3. Capacity = 10 × 3.5 = 35 kW → $\dot{m}$ = 35/1100 = **0.0318 kg/s**
4. Compressor power = 0.0318 × 200 = **6.36 kW** (check: 35/5.5 = 6.36 ✔)

Note: Subcooling — cooling the liquid from the condenser below its saturation temperature: h_4 falls, the refrigerating effect rises, work is the same → COP increases. Useful **superheating** in the evaporator increases the effect but also increases work. Lowering the evaporator pressure or raising the condenser pressure **decreases** COP.

2.3 Vapour Absorption Refrigeration (VAR) — cooling without a compressor

In VCR the compressor raises the vapour pressure and needs a lot of electricity. In VAR the same job is done cleverly: the low-pressure ammonia vapour is **absorbed** in water — pumping a liquid solution is far easier than compressing a vapour (a liquid has very small volume, so pump work is negligible). Then at high pressure the solution is **heated** to separate the ammonia again.

Components: in place of the compressor there are four things — the **absorber** (ammonia vapour dissolves in water here, heat is released so cooling water is needed), the **pump** (sends the strong solution at high pressure), the **generator** (heats the solution with steam, gas flame or solar heat to drive out ammonia vapour), and a pressure-reducing valve to return the weak solution. The condenser, expansion valve and evaporator are the same as in VCR.

When useful? Where **cheap or waste heat** is available — exhaust steam of factories, solar energy, or where there is no electricity (kerosene/gas fridges in villages — the Electrolux system). It has very few

moving parts, so it is silent and durable. **Disadvantage:** low COP (about 0.5–1).

In the **Electrolux (Servel) system** there is not even a pump — a third fluid, **hydrogen**, is added so that the total pressure is the same everywhere; in the evaporator the partial pressure of ammonia is low, so it boils at a low temperature (Dalton's law of partial pressures).

> **Note:** Summary of VCR vs VAR: VCR — **mechanical energy (work)**, high COP, noisy; VAR — **heat energy**, low COP, quiet. Refrigerant–absorbent pairs in VAR: **ammonia–water** (for sub-zero) and **water–lithium bromide** (for AC, where water is the refrigerant).

2.4 Refrigerants — how to choose and which are used

Properties of a good refrigerant: (1) **low boiling point** at normal pressure, so it boils at low temperature; (2) **high latent heat**, so a small quantity gives more cooling; (3) condenser pressure not too high and evaporator pressure slightly above atmospheric (so air does not leak in); (4) non-toxic, non-flammable and non-corrosive; (5) leaks easy to detect; (6) cheap; (7) **zero ozone depletion potential (ODP)** and **low global warming potential (GWP)**.

Main refrigerants: Ammonia (R-717) — highest latent heat, cheap, used in ice plants and cold storage; but toxic with a pungent smell (which is how leaks are detected). **R-12 (Freon)** — earlier in household fridges, now banned due to ozone depletion. **R-22** — in older ACs, being phased out. **R-134a** — ozone-safe substitute for R-12 (car AC, fridges). **R-410A, R-32** — modern ACs. **R-600a (isobutane)** — today's household fridges, natural and eco-friendly. **CO_2 (R-744)** — a natural refrigerant.

Numbering rule (R-xyz): for halocarbons x = carbon atoms − 1, y = hydrogen + 1, z = number of fluorine atoms. E.g. R-12 = R-012 → C = 1, H = 0, F = 2, remaining places chlorine = 2 → CCl_2F_2. For inorganic refrigerants R-700 + molecular weight: ammonia (17) → R-717, water (18) → R-718, CO_2 (44) → R-744.

> **Note:** Leak detection: ammonia — smell or a **sulphur candle** (white fumes); Freons — **halide torch** (flame turns green) or soap solution.

1. Key Points

- **Unit of refrigeration:** 1 ton of refrigeration (TR) = rate of freezing 1 ton (907 kg) of water at 0°C into ice in 24 hours = 211 kJ/min ≈ 3.5 kW (often taken as 210 kJ/min).
- **COP:** COP = refrigerating effect ÷ work input = Q_L/W; it can exceed 1. Carnot COP = $T_L/(T_H - T_L)$.
- **Vapour compression refrigeration (VCR):** four components — compressor (low-pressure vapour → high-pressure superheated vapour), condenser (heat rejected, vapour → liquid), expansion valve (throttling, h constant, pressure and temperature fall), evaporator (heat absorbed, liquid → vapour).
- **COP of VCR (from p–h chart):** COP = $(h_1 - h_4)/(h_2 - h_1)$, where $h_4 = h_3$ (throttling).
- **Vapour absorption refrigeration (VAR):** an absorber–generator replaces the compressor; energy is supplied as heat (steam, solar, waste heat). NH_3–H_2O (ammonia refrigerant, water absorbent) and

$LiBr–H_2O$ (water refrigerant) systems. The Electrolux refrigerator also uses hydrogen.

- **Air refrigeration (Bell–Coleman / reversed Brayton):** working fluid is air; used for aircraft cabin cooling; low COP.
- **Refrigerants:** R-12 (Freon-12, CFC – ozone depleting, banned), R-22 (HCFC), R-134a (HFC – domestic fridges/car AC), R-717 (ammonia – large cold stores, ice plants; toxic), R-744 (CO_2), R-600a (isobutane), R-290 (propane), R-718 (water).
- **Properties of a good refrigerant:** high latent heat, low boiling point, moderate condensing pressure, non-toxic, non-flammable, ozone-friendly (ODP = 0) and low GWP, cheap, easy leak detection.
- **Subcooling and superheating:** subcooling the liquid after the condenser and superheating vapour after the evaporator both increase refrigerating effect; subcooling raises COP.
- **Link to air conditioning:** psychrometry – dry bulb, wet bulb, dew point, relative humidity; comfort around 25°C DBT, 50% RH.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
COP (refrigerator)	$\mathbf{COP = Q_L/W}$
Carnot COP	$\mathbf{T_L/(T_H - T_L)}$
VCR COP	$\mathbf{(h_1 - h_4)/(h_2 - h_1)}$
Refrigerating effect	$\mathbf{R_E = h_1 - h_4}$
Compressor work	$\mathbf{W = h_2 - h_1}$
Condenser heat	$\mathbf{Q_H = h_2 - h_3}$
Throttling	$\mathbf{h_3 = h_4}$
Refrigerant mass flow	$\mathbf{m = 211 \times TR/(h_1 - h_4)\ kg/min}$
1 TR	**211 kJ/min = 3.517 kW**
HP COP	$\mathbf{COP_{HP} = COP_R + 1}$

3. Detailed Explanation

- **Working of VCR:** refrigerant boils at low pressure in the evaporator absorbing heat → compressor raises pressure and temperature → condenser rejects heat to air/water and liquefies it → expansion valve drops pressure and the cold mixture returns to the evaporator.
- **VCR vs VAR:** VCR has high COP, is electrically driven and compact; VAR has low COP, is heat driven, silent (few moving parts) and bulky — used where waste heat is available.
- **Domestic refrigerator:** hermetically sealed compressor, capillary tube expansion device, air-cooled condenser (rear grille), R-134a/R-600a.
- **Expansion devices:** capillary tube (small units), thermostatic expansion valve (TXV – AC), automatic expansion valve, float valve (large plants).

- **Improving COP:** raise evaporator temperature, lower condenser temperature, subcool; the smaller the temperature lift, the higher the COP.

4. Comparison Table — Vapour Compression vs Vapour Absorption

Basis	VCR	VAR
Energy input	Mechanical work (electricity)	Heat (steam, solar, waste)
Key device	Compressor	Absorber + generator
COP	High (3–5)	Low (≈ 0.5–1)
Noise & moving parts	More	Very few (silent)
Refrigerant	R-134a, R-22 etc.	NH_3 (water absorbent) / water (LiBr)

5. Solved Numerical Problems

Example 1. A Carnot refrigerator works between −10°C and 30°C. Find its COP.

Solution:

1. T_L = 263 K, T_H = 303 K
2. COP = 263/40 = **6.58**

Example 2. In a VCR cycle h_1 = 400, h_2 = 440, $h_3 = h_4$ = 250 kJ/kg. Find refrigerating effect, work and COP.

Solution:

1. R_E = 400 − 250 = **150 kJ/kg**
2. W = 440 − 400 = **40 kJ/kg**
3. COP = 150/40 = **3.75**

Example 3. The above plant has 5 TR capacity. Find refrigerant mass flow and compressor power.

Solution:

1. m = 211 × 5/150 = **7.03 kg/min**
2. P = 7.03 × 40/60 = **4.69 kW**

Example 4. An AC removes 3.5 kW and consumes 1 kW. Find COP and capacity in TR.

Solution:

1. COP = 3.5/1 = **3.5**
2. Capacity ≈ 3.5/3.517 ≈ **1 TR**

Example 5. Find the COP if the above Carnot refrigerator is run as a heat pump.

Solution:

1. COP_{HP} = 6.58 + 1 = **7.58**

6. Frequently Asked Facts

- 1 TR = 211 kJ/min ≈ 3.5 kW.
- Enthalpy is constant across the expansion valve in VCR ($h_3 = h_4$).
- NH_3 – ice plants/cold storage; R-134a – domestic fridges; in LiBr–H_2O water is the refrigerant.
- Air refrigeration (Bell–Coleman) – aircraft.

7. Note (Confusing Facts)

- In NH_3–H_2O ammonia is the refrigerant and water the absorbent; in LiBr–H_2O it is reversed — water is the refrigerant.
- A COP above 1 does not violate the second law.
- Heat leaves at the condenser and enters at the evaporator.

Remember: One line: compressor → condenser → expansion valve → evaporator. Assertion–Reason: (A) R-12 has been phased out. (R) It depletes the ozone layer. — Both true; R explains A.

30 Most Important One-Liner Facts — Principles of Refrigeration Plants

1. 1 TR = ?

— 211 kJ/min ≈ 3.5 kW.

2. COP?

— Refrigerating effect ÷ work.

3. Carnot refrigerator COP?

— $T_L/(T_H - T_L)$.

4. Four components of VCR?

— Compressor, condenser, expansion valve, evaporator.

5. Where is heat absorbed?

— Evaporator.

6. Where is heat rejected?

— Condenser.

7. Process in expansion valve?

— Throttling (h constant).

8. VCR COP formula?

— $(h_1 - h_4)/(h_2 - h_1)$.

9. Refrigerating effect?

— $h_1 - h_4$.

10. Energy input in VCR?

— Mechanical work.

11. Energy input in VAR?

— Heat.

12. Replaces compressor in VAR?

— Absorber and generator.

13. Refrigerant in NH_3–H_2O?

— Ammonia.

14. Refrigerant in LiBr–H_2O?

— Water.

15. Third gas in Electrolux?

— Hydrogen.

16. Aircraft cabin cooling?

— Air refrigeration (Bell–Coleman).

17. R-717?

— Ammonia.

18. R-744?

— CO_2.

19. R-718?

— Water.

20. Domestic fridge refrigerant?

— R-134a / R-600a.

21. Ozone-depleting refrigerant?

— R-12 (CFC).

22. Ice plant refrigerant?

— Ammonia.

23. Expansion device in fridge?

— Capillary tube.

24. Expansion device in AC?

— Thermostatic expansion valve.

25. Effect of subcooling?

— COP increases.

26. Feature of VAR?

— Silent, heat driven.

27. Refrigerant state after compressor?

— High-pressure superheated vapour.

28. COP with higher evaporator temp?

— Increases.

29. Comfort DBT and RH?

— ≈ 25°C, 50%.

30. Refrigeration is based on?

— Second law (Clausius).

Practice MCQs — Principles of Refrigeration Plants

Q1. 1 ton of refrigeration equals:

(a) 21 kJ/min
(b) 211 kJ/min
(c) 2110 kJ/min
(d) 3.5 kJ/min

✔ **Answer: (b) 211 kJ/min**

Explanation: ≈ 3.5 kW.

Q2. In VCR the refrigerant absorbs heat in the:

(a) Condenser
(b) Evaporator
(c) Compressor
(d) Expansion valve

✔ **Answer: (b) Evaporator**

Explanation: Evaporator.

Q3. Process in the expansion valve:

(a) Isentropic
(b) Isenthalpic (throttling)
(c) Isothermal
(d) Isobaric

✔ **Answer: (b) Isenthalpic (throttling)**

Explanation: h constant.

Q4. In VAR the compressor is replaced by:

(a) Pump
(b) Absorber and generator
(c) Turbine
(d) Nozzle

✔ **Answer: (b) Absorber and generator**

Explanation: Thermal compression.

Q5. Refrigerant in a LiBr–water system:

(a) LiBr
(b) Water
(c) Ammonia
(d) Hydrogen

✔ **Answer: (b) Water**

Explanation: LiBr is the absorbent.

Q6. Used for aircraft air conditioning:

(a) VCR
(b) VAR
(c) Air refrigeration
(d) Steam jet

✔ **Answer: (c) Air refrigeration**

Explanation: Light, air available.

Q7. $h_1 = 400$, $h_2 = 440$, $h_4 = 250$ — COP:

(a) 3.75
(b) 4.75
(c) 2.5
(d) 6

✔ **Answer: (a) 3.75**

Explanation: 150/40.

Q8. Carnot refrigerator COP between −10°C and 30°C:

(a) 0.33
(b) 6.58
(c) 7.58
(d) 10

✔ **Answer: (b) 6.58**

Explanation: 263/40.

Q9. R-717 is:

(a) Freon
(b) Ammonia
(c) CO_2
(d) Water

✔ **Answer: (b) Ammonia**

Explanation: Ammonia.

Q10. Expansion device in a domestic refrigerator:

(a) Capillary tube
(b) Float valve
(c) TXV
(d) Nozzle

✔ **Answer: (a) Capillary tube**

Explanation: Simple, cheap.

Q11. Most damage to the ozone layer is by:

(a) R-134a
(b) R-12
(c) R-717
(d) R-744

✔ **Answer: (b) R-12**

Explanation: CFC.

Q12. Which is not a desirable refrigerant property?

(a) High latent heat
(b) Toxicity
(c) Non-flammable
(d) Low boiling point

✔ **Answer: (b) Toxicity**

Explanation: Should be non-toxic.

Q13. State of refrigerant after the condenser in VCR:

(a) High-pressure liquid
(b) Low-pressure vapour
(c) High-pressure vapour
(d) Solid

✔ **Answer: (a) High-pressure liquid**

Explanation: Condensed liquid.

Q14. If $COP_R = 4$, COP_{HP} is:

(a) 3
(b) 4
(c) 5
(d) 0.25

✔ **Answer: (c) 5**

Explanation: 4 + 1.

Q15. Raising evaporator temperature makes COP:

(a) Decrease
(b) Increase
(c) Constant
(d) Zero

✔ **Answer: (b) Increase**

Explanation: Temperature lift reduces.

SSC JE / UPSSSC JE Previous Year Questions — Principles of Refrigeration Plants

Q1. One ton of refrigeration is equal to:

SSC JE (Previous Year)

(a) 210 kJ/min
(b) 21 kJ/min
(c) 2100 kJ/min
(d) 1 kJ/s

✔ **Answer: (a) 210 kJ/min**

Explanation: ≈ 210–211 kJ/min.

Q2. In VCR, the refrigerant rejects heat in the:

SSC JE (Previous Year)

(a) Evaporator
(b) Condenser
(c) Expansion valve
(d) Compressor

✔ **Answer: (b) Condenser**

Explanation: Condenser.

Q3. Absorbent in the ammonia absorption system: SSC JE (Previous Year)

(a) Ammonia
(b) Water
(c) LiBr
(d) Hydrogen

✔ **Answer: (b) Water**

Explanation: Water.

Q4. Bell–Coleman cycle is a: SSC JE (Previous Year)

(a) Reversed Carnot
(b) Reversed Brayton (Joule)
(c) Otto
(d) Rankine

✔ **Answer: (b) Reversed Brayton (Joule)**

Explanation: Air refrigeration.

Q5. COP of a refrigerator: SSC JE (Previous Year)

(a) Always less than 1
(b) Can be more than 1
(c) Always 1
(d) Negative

✔ **Answer: (b) Can be more than 1**

Explanation: $Q_L > W$ is possible.

Q6. In the expansion valve, constant is: SSC JE (Previous Year)

(a) Entropy
(b) Enthalpy
(c) Pressure
(d) Temperature

✔ **Answer: (b) Enthalpy**

Explanation: Throttling.

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